

STAT 250 FINAL PROJECT REPORT

Exploring the Relationship Between Lifestyle Habits, Family History and Obesity in Individuals

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ABSTRACT

Our project analyzes obesity levels' relationship with certain lifestyle factors such as dietary habits, daily habits and obesity levels' relationship with each individuals family history of obesity. R and Python were used to conduct all the analysis and to create neccassary graphs and figures to visualize our data for the report. First we did an exploratory data analysis as its an essential step to have an overview understanding of the structure of our data. Then we made simple graphs to interpret relationships between variables visually. Finally to analyze and conclude findings regarding the relationship between variables, we performed tests.

INTRODUCTION

Obesity is a condition shaped by various lifestyle habits, genetics and individuals' behaviour. According to WHO in 2019 alone, high BMI levels were the reason of 5 million deaths. This highlights the need of researching obesity and the factors that lead to this condition. In this project, we aim to investigate the relationship between various demographic and lifestyle characteristics and obesity levels using a real-world dataset.

METHODOLOGY

1. Dateset Overview

This data set is called "Estimation of Obesity Levels Based On Eating Habits and Physical Condition" and it is taken from UC Irvine Machine Learning Repository. This dataset include data for the estimation of obesity levels in individuals from the countries of Mexico, Peru and Colombia, based on their eating habits and physical condition. The data contains 17 variables and 2111 records.

Variable Name	Type	Question	Possible Answers
Gender	Categorical	What is your gender?	- Female - Male
Age	Continuous	What is your age?	Numeric value
Height	Continuous	What is your height?	Numeric value in meters
Weight	Continuous	What is your weight?	Numeric value in kilograms

family_history_with_overweight	Binary	Has a family member suffered or suffers from overweight?	-Yes -No
FAVC	Binary	Do you eat high caloric food frequently?	-Yes -No
FCVC	Integer	Do you usually eat vegetables in your meals?	-Never -Sometimes -Always
NCP	Continuous	How many main meals do you have daily?	Numeric value
CAEC	Categorical	Do you eat any food between meals?	-No -Sometimes -Frequently -Always
SMOKE	Binary	Do you smoke?	-Yes -No
CH2O	Continuous	How much water do you drink daily?	Numeric value in liters
SCC	Binary	Do you monitor the calories you eat daily?	-Yes -No
FAF	Continuous	How often do you have physical activity?	Numeric value in days
TUE	Integer	How much time do you use technological devices such as cell phone, videogames, television, computer and others?	Numeric value in hours
CALC	Categorical	How often do you drink alcohol?	-I do not drink -Sometimes -Frequently -Always
		Which transportation	-Automobile -Motorbike

MTRANS	Categorical	do you usually use?	-Bike Public -Transportation -Walking
NObeyesdad	Categorical	Obesity level	-Insufficient_Weight -Normal_Weight -Overweight_Level_I -Overweight_Level_II -Obesity_Type_I -Obesity_Type_II -Obesity_Type_III

Table 1 “Dataset overview”

2. Descriptive Statistics

Statistic	Age	Height	Weight	FCVC	NCP	CH2O	FAF	TUE
Minimum	14	1.45	39	1	1	1	0	0
1st Quartile	19.95	1.63	65.47	2	2.65	1.58	0.12	0
Median	22	1.70	83	2.38	3	2	1	0.63
Mean	24.31	1.70	86.59	2.41	2.68	2	1.01	0.65
Standard Dev.	6.34	0.093	26.19	0.53	0.77	0.61	0.85	0.61
3rd Quartile	26	1.76	107.43	3	3	2.47	1.66	1
Maximum	61	1.98	173	3	4	3	3	2

Table 2 “Descriptive statistics of numerical variables”

Table 2 refers to our analysis with descriptive statistics to understand the distribution of the dataset’s numerical variables. The age of participants ranges from 14 to 61 years, with a mean of 24.31 and a median of 22 suggesting a slightly right-skewed distribution concentrated among younger individuals. The height variable is fairly symmetrically distributed, with a mean of 1.70 meters and a standard deviation of 0.093, indicating limited variability. Weight shows considerable spread, ranging from 39 to 173 kg, with a mean of 86.59 and a standard deviation of 26.19, reflecting the inclusion of individuals from various obesity categories. Nutritional and lifestyle variables also exhibit meaningful variation. For instance, FCVC (frequency of vegetable consumption) and CH2O (water intake) both cluster around 2 to 3, suggesting relatively healthy eating habits, while FAF (physical activity frequency) and TUE (technology usage) show more variability, with medians of 1 and 0.63 respectively, and standard deviations around 0.85 and 0.61. These summaries provide important insight into the dataset and guided our choice of methods for further statistical analysis.

ANALYSIS AND RESULTS

Association Between Alcohol Consumption Frequency and Obesity Levels

Null Hypothesis:

H_0 : Alcohol consumption frequency is independent of obesity.

Alternative Hypothesis:

H_1 : Alcohol consumption frequency is associated with obesity.

CALC: Individuals were asked how frequently they consumed alcohol with the answers being give as: no, Sometimes, Frequently, Always (Categorical).

As mentioned previously, obesity is also categorical (7 levels) Since both variables are categorical we used a chi-square test of independence to determine if there is a significant association between obesity levels and alcohol consumption frequency.

Using software code we obtain the chi-square statistic to be 338.58 with a df of 18 resulting in a p-value of 5.29×10^{-61}

With this extremely small p-value we reject H_0 . This means that there is a significant association between obesity level and alcohol consumption frequency.

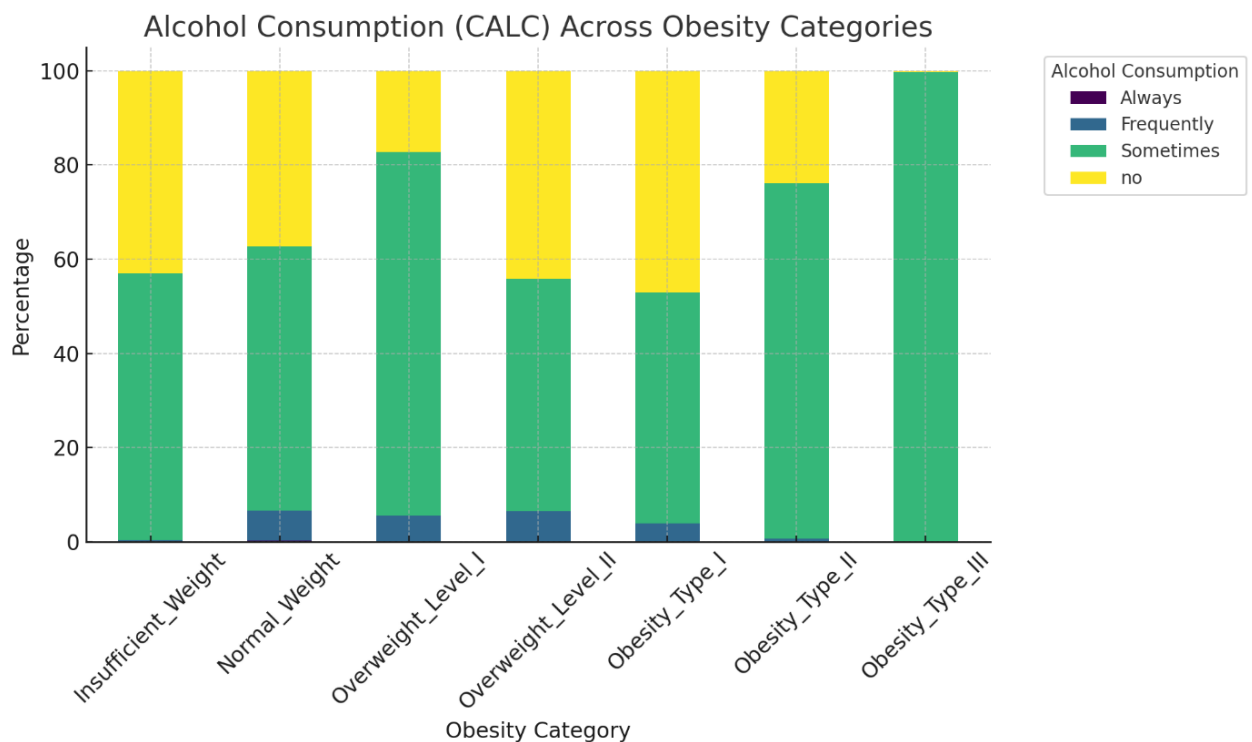


Figure 1

To further investigate and visualize the relationship, Figure 1 shows the distribution of alcohol consumption across obesity levels. While the alcohol consumption patterns vary overall, we can see that individuals that are Type 2 and Type 3 level obese have higher and more consistent amount of alcohol consumption in comparison to lower obesity levels.

Calorie Tracking Habits Across Different Obesity Categories

Null Hypothesis:

H₀: Calorie monitoring (SCC) is independent of obesity level.

Alternative Hypothesis:

H₁: There is a significant association between obesity level and whether individuals monitor their calories.

A chi-square test of independence was done to conclude whether there is an association between obesity levels and whether individuals track their calorie intake (SCC). The test resulted in a **chi-square statistic of 123.02 which yields a p-value < 0.01**. This results in a rejection of the null hypothesis meaning that calorie monitoring and obesity levels are not independent of each other.

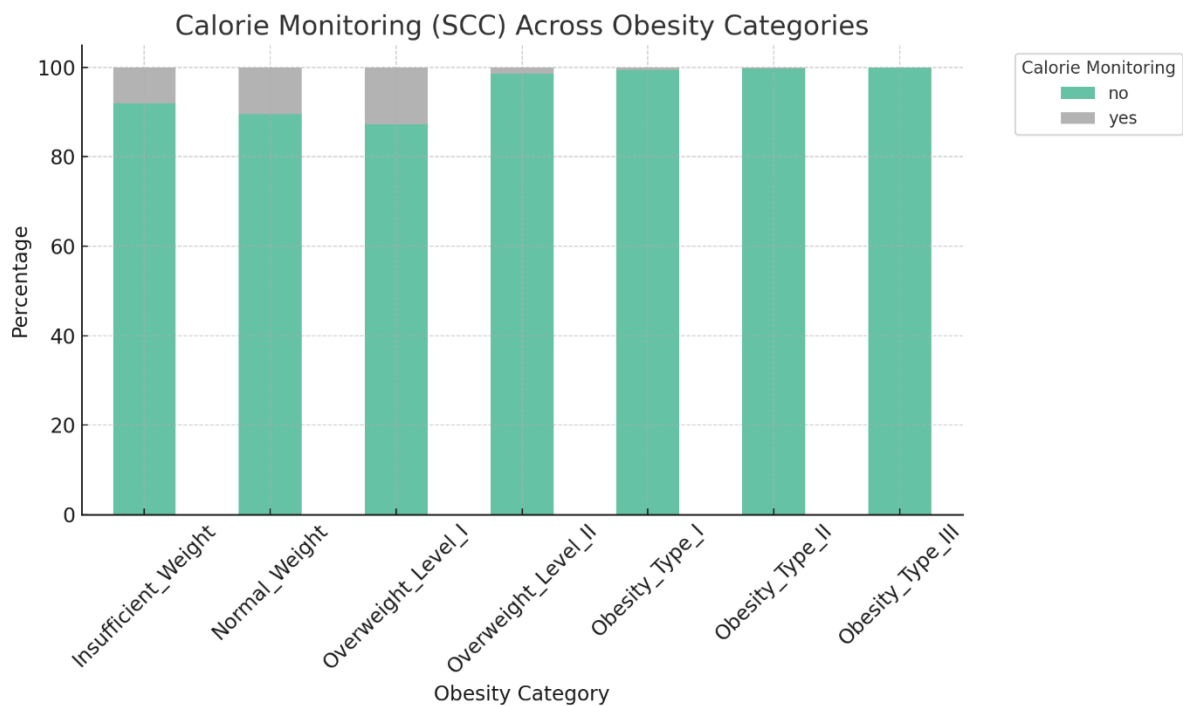


Figure 2

Figure 2 shows the distribution of calorie monitoring behavior (SCC) among obesity levels. While the differences are not too drastic, the graph shows that individuals who are labeled as Insufficient Weight, Normal Weight and Overweight Level 1 have a higher tendency of monitoring their calories in comparison to the categories above them. Although the percentages are not too drastically different since our sample size is large (2,000) it results in a significant difference. This suggests that calorie tracking behaviour might results in lower obesity rates.

To further back our findings, A study by the University of Florida's College of Public Health and Health Professions (2024) conducted a study on 75 adults with overweight or obesity and found that monitoring caloric intake at least three days a week results in long term weight loss maintenance and that monitoring caloric intake 5-6 times a week results in even greater weight loss. In conclusion the research emphasizes that monitoring caloric intake is crucial for weight loss and preventing weight gain.

The Link Between Obesity Levels and Physical Activity Frequency

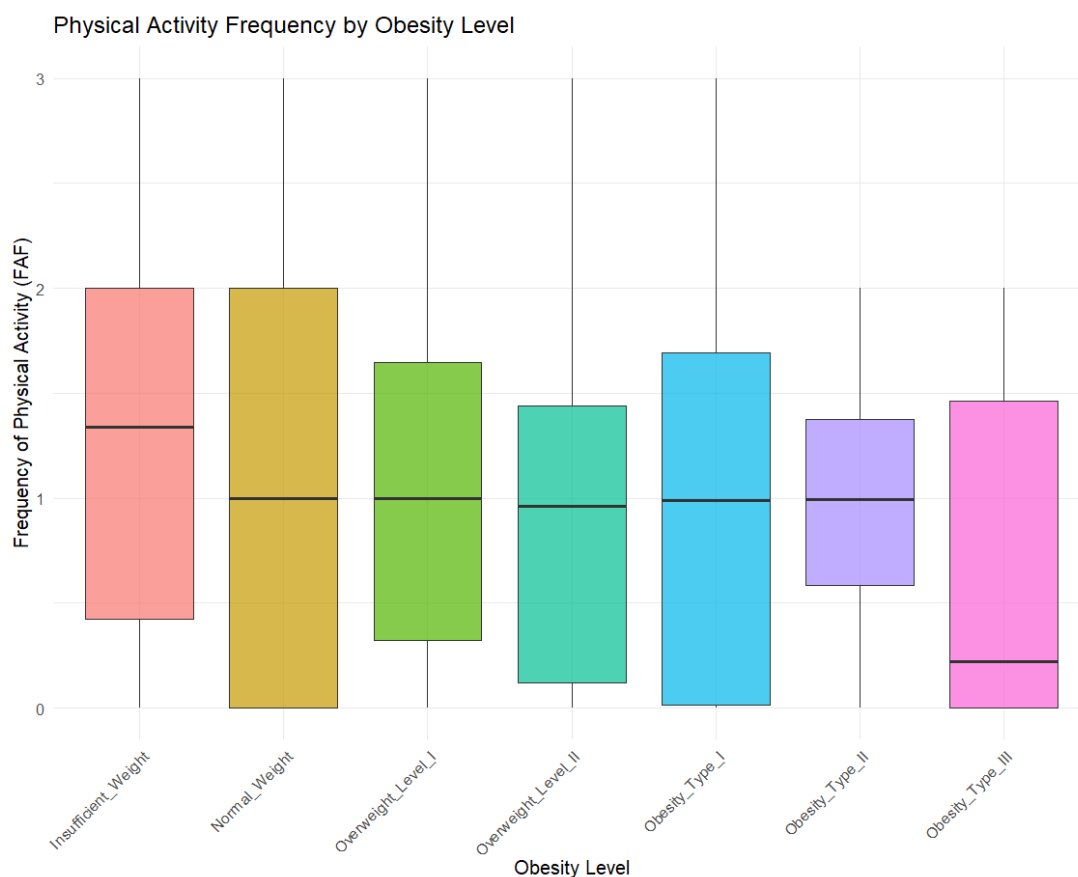


Figure 3

In this study, Frequency of Physical Activity (FAF) refers to how many days per week an individual engages in physical activity. We aimed to investigate whether individuals who are more physically active tend to have lower obesity levels. According to the boxplot that displays the distribution of physical activity frequency (FAF) across various obesity levels people in lower obesity categories (such as Insufficient Weight and Normal Weight) typically have higher median physical activity frequencies than people in higher obesity categories (such as Obesity Type II and III). As obesity levels rise, the median frequency of physical activity tends to fall, with lower levels of physical activity observed in more severely obese groups.

To statistically examine whether physical activity frequency differs across obesity levels, a one-way ANOVA was performed. The dependent variable was FAF, and the independent variable was Obesity Level, which included seven groups ranging from Insufficient Weight to Obesity Type III.

	Df	Sum Sq	Mean Sq	F value	P value
NObesidad	6	72.5	12.084	17.48	<2e-16
Residuals	2104	1454.1	0.691		
Total	2110				

Table 3 “Anova table”

To investigate whether these differences are statistically meaningful, we set up two hypotheses.

Null Hypothesis (H_0): The mean frequency of physical activity is equal across all obesity levels.

Alternative hypothesis (H_1): At least one obesity level group has a mean frequency of physical activity that differs from the others.

To test these hypotheses, we conducted a one-way Analysis of Variance (ANOVA), which allows us to compare the means of more than two groups. The ANOVA results showed a significant difference in physical activity frequency across the seven obesity levels, with an F value of 17.48 and a p-value less than 0.001. Since the p-value is well below 0.05, we rejected the null hypothesis. This means there is strong evidence that physical activity frequency varies between different obesity levels.

To pinpoint which groups differ, we performed Tukey's post-hoc test. This analysis confirmed that individuals in lower obesity categories generally have higher physical activity frequencies compared to those in higher obesity categories.

However, before fully trusting the ANOVA results, we checked the key assumptions underlying this test. The ANOVA requires that the data within each group be approximately normally distributed and that the variances across groups be similar. Unfortunately, the Shapiro-Wilk test showed the residuals were not normally distributed, and Levene's test revealed significant differences in variances between groups. These violations mean the ANOVA results should be interpreted with caution.

To overcome this issue, we used the Kruskal-Wallis test, a non-parametric alternative that doesn't assume normality or equal variances. The Kruskal-Wallis test also indicated significant differences in physical activity frequency across obesity groups ($\chi^2(6) = 88.42$, $p < 0.001$), supporting the ANOVA findings. Post-hoc pairwise comparisons following the Kruskal-Wallis test confirmed that many obesity level groups differ significantly in their physical activity frequencies.

In summary, the analysis suggests a clear association: individuals with lower obesity levels tend to engage in physical activity more frequently, while those with higher obesity levels tend to be less active. Overall, these findings highlight the importance of physical activity in relation to obesity and suggest that encouraging regular physical activity might be a key factor in managing or preventing obesity.

A Statistical Inference on Smoking Rates Across Obesity Groups

We began our analysis by exploring the dataset and understanding the key variables:

ObeseGroup: A binary variable we created based on the original NObeyesdad variable. Individuals classified as "Obesity_Type_I", "Obesity_Type_II", or "Obesity_Type_III" were labeled as "Obese", while all others were labeled as "Non-Obese".

SMOKE: A categorical variable indicating whether the individual smokes ("yes") or does not smoke ("no").

We examined the frequencies of these variables: There were 1139 Non-Obese and 972 Obese individuals in the dataset. A total of 44 individuals were smokers, and 2067 were non-smokers. The contingency table showed the following breakdown:

Obese Groups	No-(Non-smoker)	Yes (Smoker)
Non-Obese	1117	22
Obese	950	22

Table 4

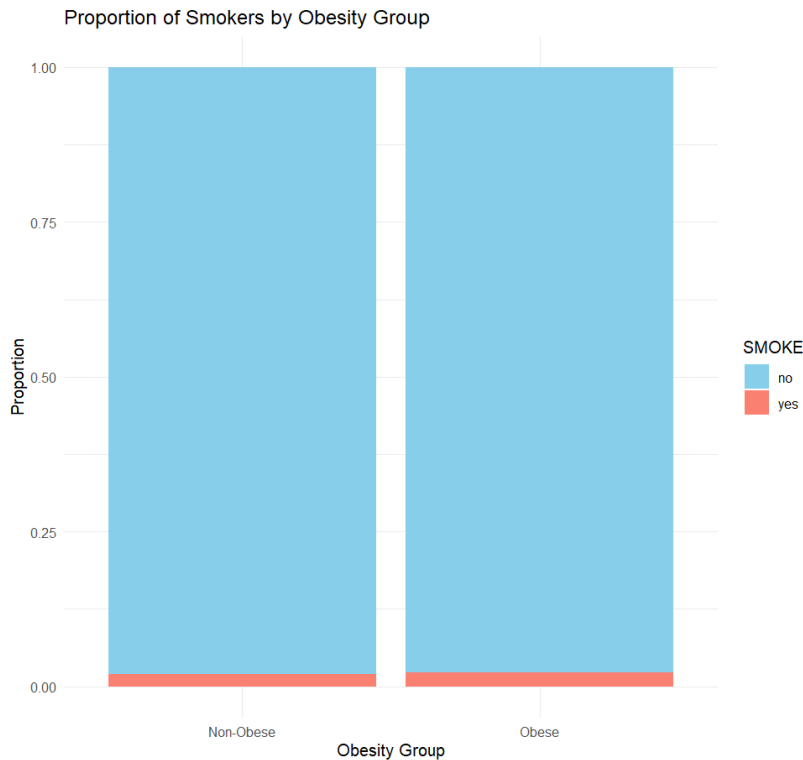


Figure 3

We applied a two-sample test for equality of proportions to compare the smoking rates in the two groups.

Before conducting the two-sample proportion test, we verified that the necessary statistical assumptions were met. First, the outcome variable — smoking status — is binary, with two clearly defined categories: "yes" (smoker) and "no" (non-smoker). Second, the two groups being compared — Obese and Non-Obese individuals — are independent of each other, meaning that no individual belongs to both groups simultaneously. Finally, both groups have sufficiently large sample sizes (greater than 30), which ensures the validity of the normal approximation used in the proportion test. Given that all assumptions were satisfied, we proceeded with the two-sample test for equality of proportions.

Null Hypothesis (H_0): The proportion of smokers is equal in both groups.

Alternative Hypothesis (H_1): The proportions are different.

The R function `prop.test()` produced the following output:

-p-value = 0.7046

-95% Confidence Interval: [-0.0099, 0.0166]

Proportion of smokers:

-Obese group: $22 / 972 \approx 2.26\%$

-Non-Obese group: $22 / 1139 \approx 1.93\%$

Since the p-value is much greater than the standard alpha level of 0.05, we fail to reject the null hypothesis. Additionally, the confidence interval for the difference in proportions includes zero, providing further evidence that the observed difference is not statistically significant.

This analysis shows that there is no statistically significant difference in the proportion of smokers between obese and non-obese individuals in the dataset. Although the obese group had a slightly higher smoking rate, the difference is not meaningful from a statistical perspective. This suggests that smoking behavior does not significantly differ based on obesity status in this sample.

CONCLUSION

This study provides a comprehensive statistical examination of the relationship between obesity levels and various lifestyle factors, including dietary habits, physical activity, alcohol consumption, calorie monitoring, and smoking. Our analysis revealed that both alcohol consumption frequency and calorie monitoring behavior are significantly associated with obesity levels. Individuals who monitor their caloric intake more frequently tend to belong to lower obesity categories, highlighting the importance of self-awareness in dietary habits. Additionally, we observed a strong inverse relationship between physical activity frequency and obesity, further supported by both ANOVA and non-parametric tests. These results reinforce the role of active lifestyles in weight management. On the other hand, smoking behavior did not significantly differ across obesity groups, suggesting that smoking may not be a key distinguishing factor in obesity status within this dataset.

It is also important to note that the dataset used in this study includes individuals from Mexico, Peru, and Colombia. These countries share certain socio-economic and cultural similarities with Turkey, such as urbanization trends, dietary transitions, and increasing sedentary lifestyles. While direct generalizations should be made cautiously, the insights gained from this dataset can still be valuable for understanding and addressing obesity-related behaviors in the Turkish context.

Overall, our findings underscore the importance of promoting healthier eating, regular exercise, and behavioral monitoring as strategies to reduce obesity. While the results are insightful, further research with more robust experimental designs and localized data may be beneficial.

References

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